

PROBLEM & EVIDENCE

Problem:

Students need part-time work compatible with their studies.
Existing job listings may not consider individual constraints.

Unsuitable jobs can cause:

- Schedule conflicts
- Excessive working hours
- Long travel distance
- Skill mismatch

Our Solution:

Student Profile + Job Dataset → Constraint Filtering → A* Search → Best Job

Target Users:

🎓 University students seeking suitable part-time employment

Key Outcome:

Recommend a valid job based on constraints + suitability cost



PROBLEM FORMULATION

Formal Search Formulation

Initial State:

Student
Jobs

Hard Constraints:

- Age
- Skills
- Availability
- Maximum hours
- Maximum distance

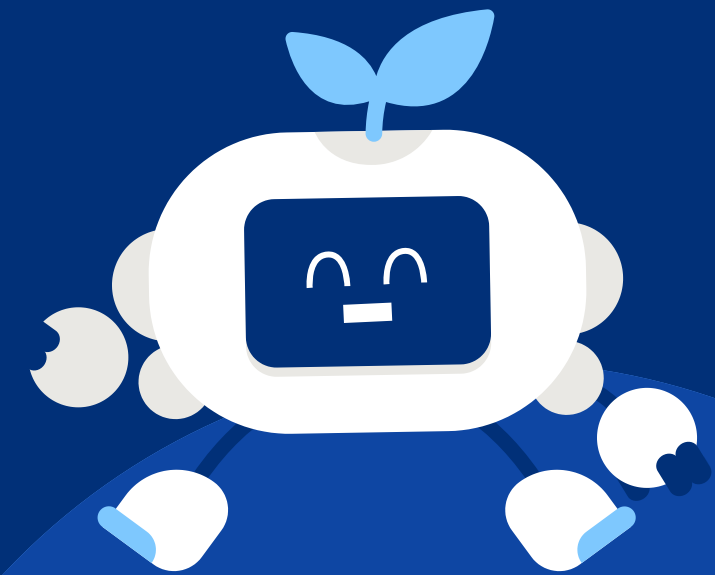
Actions:

1. Select job
2. Check constraints
3. Reject invalid job
4. Calculate $g(n)$
5. Calculate $h(n)$
6. Calculate $f(n)$

A* Evaluation:

$$f(n) = g(n) + h(n)$$

- $g(n)$ = current suitability cost
- $h(n)$ = estimated suitability cost
- Lowest $f(n)$ = preferred candidate

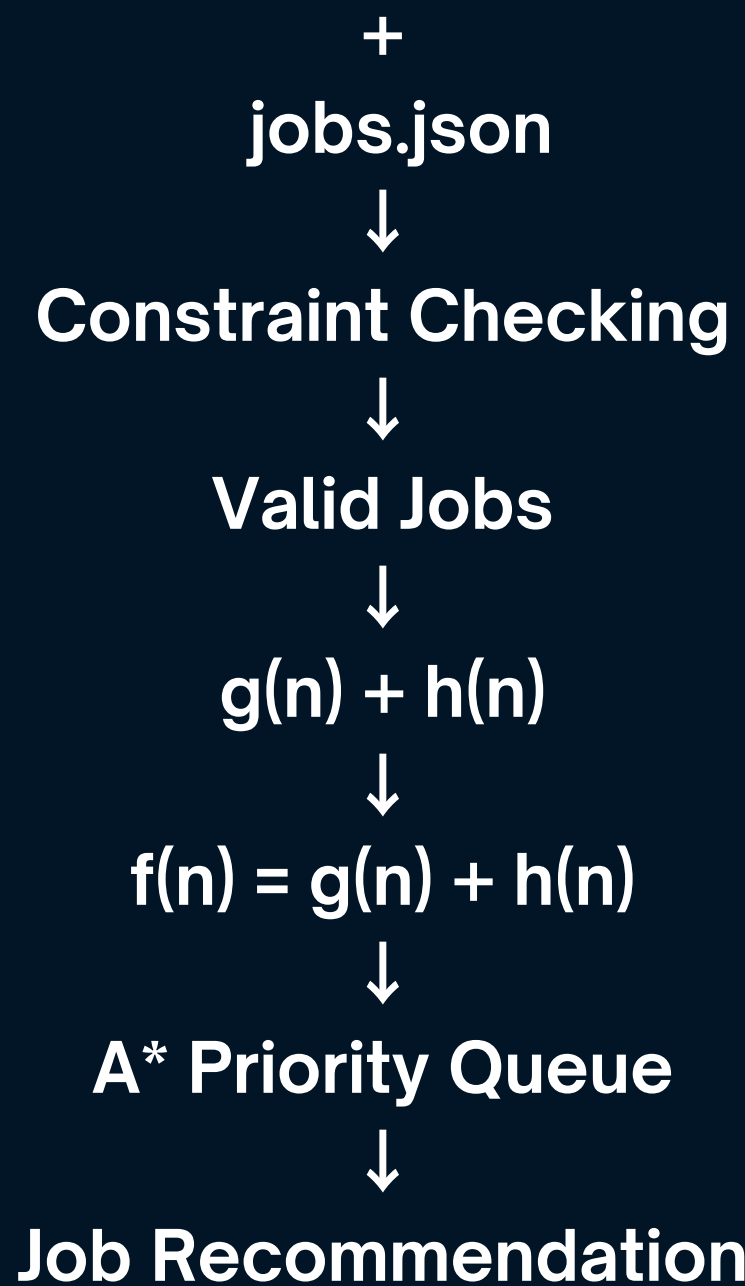


IMPLEMENTATION

Modular Design:

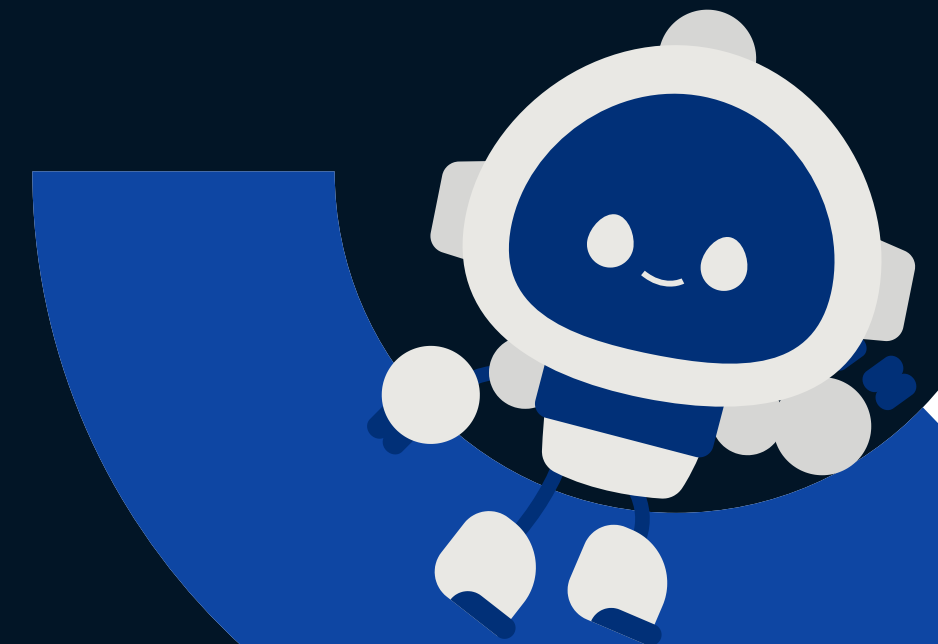
File	Responsibility
main.py	No optimisation
fairwork_ai.py	May return poor option
heuristics.py	Can ignore overall cost
jobs.json	Job dataset
test_fairwork_ai.py	Automated testing

Python Prototype: Student Profile



Search Metrics:

- Search cost
- Jobs expanded
- Execution time



LIVE DEMONSTRATION & TESTING

The screenshot displays a Visual Studio Code editor window with the following components:

- Explorer (Left Panel):** Shows the project structure for 'BCS2143_BIT3203_AI_Changemaker'. The 'data' folder is expanded, showing 'jobs.json' (modified) and 'README.md'. Other folders like 'docs', 'presentation', 'results', and 'src' are also visible.
- Main Editor:** Displays the content of 'jobs.json', which is a JSON array of three job objects. The cursor is at line 30, column 2.
- Chat (Right Panel):** Shows a 'Build with Agent' section with a message: 'AI responses may be inaccurate. Generate Agent Instructions to add to your codebase.' Below this is a tip: 'Tip: Use /create-agent to scaffold an agent for your workflow.' and a button to '+ jobs.json'.
- Status Bar (Bottom):** Shows the current file is 'jobs.json' (JSON), the cursor is at 'Ln 301, Col 2', and the encoding is 'UTF-8'.

jobs.json Content:

```
[
  {
    "id": "J001",
    "title": "Cafe Crew",
    "company": "Campus Cafe",
    "min_age": 18,
    "skills": [
      "Customer Service",
      "Communication"
    ],
    "distance": 2.0,
    "hours": 10,
    "salary": 12.0,
    "shift_start": 18,
    "shift_end": 22
  },
  {
    "id": "J002",
    "title": "Retail Assistant",
    "company": "Student Mall",
    "min_age": 18,
    "skills": [
      "Customer Service",
      "Sales"
    ],
    "distance": 4.0,
    "hours": 15,
    "salary": 11.5,
    "shift_start": 17,
    "shift_end": 21
  },
  {
    "id": "J003",
    "title": "Data Entry Assistant",
    "company": "Digital Services",
    "min_age": 18,
```

RESPONSIBLE AI

Principle	FairWork AI Mitigation
Fairness	Consistent constraint rules
Privacy	Minimise personal data
Transparency	Explain recommendation factors
Safety	Reject unsuitable jobs
Accessibility	Designed around student constraints
Security	Protect profile/job data
Sustainability	Lightweight local search

SDG IMPACT

SDG 8: Decent Work and Economic Growth

Improves access to appropriate employment

Considers age, working hours and job suitability

